

Dual-Stage Deep Learning for Explainable Retinal Disease Diagnosis via OCT Segmentation and Fluid Biomarker Classification

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Abstract—Retinal disorders, including diabetic retinopathy and age-related macular degeneration, collectively impact hundreds of millions globally, straining healthcare systems and contributing to clinician burnout. To address this challenge, we propose a dual-stage deep learning framework for automated, interpretable diagnosis of retinal pathologies from Optical Coherence Tomography (OCT) scans. The pipeline first segments clinically relevant fluid biomarkers—intraretinal fluid (IRF), subretinal fluid (SRF), and pigment epithelial detachments (PED)—to generate explainable disease proxies. These fluid maps are then integrated with raw OCT scans to classify pathologies: Choroidal Neo-Vascularization, Drusen, and Diabetic Macular Edema.

The segmentation model, leveraging a DeepLabV3 architecture, achieved state-of-the-art performance with a Dice score of 95% and a Jaccard index of 90%, providing pixel-level interpretability of fluid regions. For classification, a hybrid network combining fluid maps and OCT scans via transfer learning (ResNet-50 backbone) attained 94% accuracy, 93% precision, and 92% F1-score, outperforming baseline models trained solely on raw scans. By explicitly linking fluid biomarkers to disease phenotypes, our approach bridges the gap between opaque ‘black-box’ AI and clinical reasoning, offering ophthalmologists actionable visual evidence to support diagnosis.

This work demonstrates how explainable AI can enhance diagnostic transparency while maintaining high accuracy, potentially reducing workloads and improving trust in automated systems. The code and preprocessed datasets are made publicly available to encourage reproducibility and clinical adoption.

Index Terms—Segmentation, Classification, Optical Coherence Tomography

I. INTRODUCTION

The advent of deep learning has brought transformative changes across various fields, with healthcare emerging as one of its most impactful beneficiaries. From enabling genome analysis for personalized treatment strategies to automating the generation of medical reports [1], deep learning technologies are streamlining workflows and allowing clinicians to focus on complex, high-priority cases. Moreover, deep learning models are increasingly being employed to support diagnostic processes across diverse medical imaging modalities. By identifying abnormal patterns and providing reliable reference points, these systems can accelerate diagnosis and assist in crafting

timely treatment plans—an especially critical advantage for rapidly progressing conditions.

In this study, we propose a deep learning-based framework to automate the diagnosis of prevalent retinal diseases, aiming to support ophthalmologists in managing the growing volume of patient cases more efficiently. By alleviating the burden of routine analysis, this approach allows clinicians to dedicate more attention to critical cases. The architecture is structured as a two-stage pipeline comprising a segmentation module followed by a classification module. The segmentation model is responsible for detecting key retinal biomarkers, including intra-retinal fluid (IRF), sub-retinal fluid (SRF), pigment epithelial detachment (PED), disruptions in the inner/outer segment (IS/OS) integrity, and subretinal hyper-reflective material (SHRM). These segmented features are subsequently passed to a classifier, which predicts the presence of one of three conditions: Drusen [2], Diabetic Macular Edema (DME) [3], Choroidal Neovascularization (CNV) [4], or a Normal retina. This modular design leverages well-documented clinical correlations between fluid accumulation and structural anomalies with specific retinal pathologies, enhancing interpretability and diagnostic reliability.

The proposed work contributes to the field by:

- Integration of explainable fluid segmentation with disease classification.
- Clinical relevance through transparency and efficiency gains.
- Computer-Assisted Diagnosis for common retinal disorders.

The rest of the paper is structured as follows: Section II is a literature review, Section III describes the methodology, Section IV lays out the results of the study. Section V discusses the implications and limitations of the proposed model and Section VI is the conclusion followed by the sources cited in the paper.

II. RELATED WORK

A. Retinal Disease Classification

[5] proposed deep learning models that categorize patients’ OCT images into four categories such as Choroidal neovas-

ularization (CNV), Diabetic macular edema (DME), Drusen, and Normal. Two different models were proposed. One using three binary Convolutional Neural Network (CNN) classifiers and the other using four binary CNN classifiers. Several CNNs, such as VGG16, VGG19, ResNet50, ResNet152, DenseNet121, and InceptionV3, are adapted as feature extractors to develop the binary classifiers. Among them, the proposed model using VGG16 for CNV vs. Other classes, VGG16 for DME vs. other classes, VGG19 for Drusen vs. Other classes, and InceptionV3 for Normal vs. other classes.

B. Retinal OCT Segmentation

[6] proposed automated segmentation of retinal layers in OCT images which evolved from traditional methods like graph-based algorithms and active contours, which struggled with pathological distortions, to deep learning (DL) approaches such as U-Net and fully convolutional networks (FCNs) that leverage hierarchical feature learning for sub-pixel accuracy. While recent advances integrate attention mechanisms and hybrid architectures to improve robustness, challenges persist in early-stage diseases like mild diabetic retinopathy, where minimal structural changes and inter-grader variability degrade performance. Post-processing techniques, including graph-cut optimization and shape priors, have been proposed to refine CNN outputs, yet few studies address mild retinopathy or explicitly model inter-operator annotation differences. The proposed FCN-Gaussian process (GP) framework advances this field by combining FCNs' localization capabilities with GP regression's spatial uncertainty modeling, enhancing anatomical coherence in ambiguous boundaries. This hybrid approach addresses limitations of standalone CNNs, which often exhibit mean unsigned errors of 3–5 pixels in mild cases, and aligns with trends toward probabilistic refinement in medical imaging.

III. METHODOLOGY

A. Datasets

1) *Classification Dataset*: The dataset used for training the classification model was sourced from Mendeley. The dataset was constructed by [7]. The dataset comprises of over 200,000 retinal OCT scans in jpg format for 4 classes: CNV, DME, Drusen And Normal. The images in the dataset are already augmented with random noise like shearing, cropping, translation and rotation, making it a robust dataset to train models with. However, the images had a lot of salt and pepper noise, to rectify this, a median filter was utilized to ensure preservation of details while eliminating the salt and pepper noise.

2) *Segmentation Dataset*: The dataset used for training the segmentation model was developed by [8], comprising of 3049 images along with corresponding segmentation masks highlighting fluid regions, which were annotated by junior ophthalmologists and verified/corrected by senior ophthalmologists. The annotated regions include fluids and structural abnormalities. However a major problem with the dataset was the non-uniformity of the segmentation mask pixels, i.e. masks

corresponding to red weren't precisely (255, 0, 0) but had variances. To mitigate this, K-Means clustering was utilized to group similar pixel intensities into the specified class indices which was then used to train the model.

B. Model Architectures

This section details the architectures and implementation specifics of the classification and segmentation models developed for retinal OCT analysis.

1) *Classification Model*: The classification task employs a VGG16 [9] architecture to categorize OCT images into four classes:

- *Choroidal Neovascularization (CNV)*
- *Diabetic Macular Edema (DME)*
- *Drusen*
- *Normal*

Key specifications:

- *Input*: Grayscale OCT images (224×224 pixels)
- *Backbone*: ImageNet-pretrained weights with modified final FC layer
- *Training protocol*:
 - Optimizer: Adam ($\beta_1 = 0.9, \beta_2 = 0.999$)
 - Initial Learning rate: 1×10^{-4}
 - Loss: Categorical cross-entropy with label smoothing ($\alpha = 0.1$)
 - Augmentation: Horizontal flips, $\pm 15^\circ$ rotation, $\pm 20\%$ intensity scaling

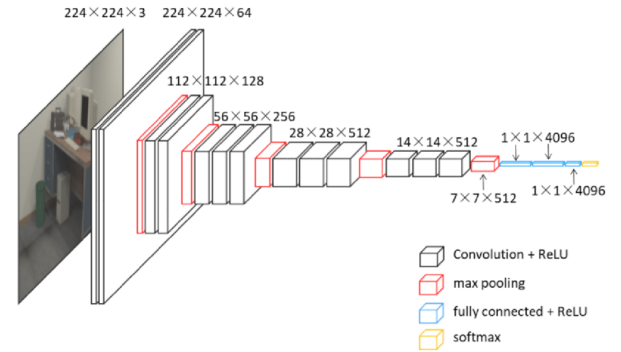


Fig. 1. Visualization of VGG16 architecture used for classification tasks.

2) *Segmentation Model*: The semantic segmentation task uses *DeepLabV3* with a ResNet-50 backbone, which is shown in Fig 2., for pixel-wise retinal layer boundary detection.

Architecture details:

- *Input*: 512×512 normalized OCT scans
- *Output*: 5-class pixel masks
- *Training configuration*:
 - Loss: Categorical Cross-Entropy Loss
 - Optimizer: Adam
 - Learning schedule: Reduce LR On Plateau

Evaluation metrics:

- Mean Intersection Over Union
- Dice similarity coefficient (DSC)
- Jaccard Index

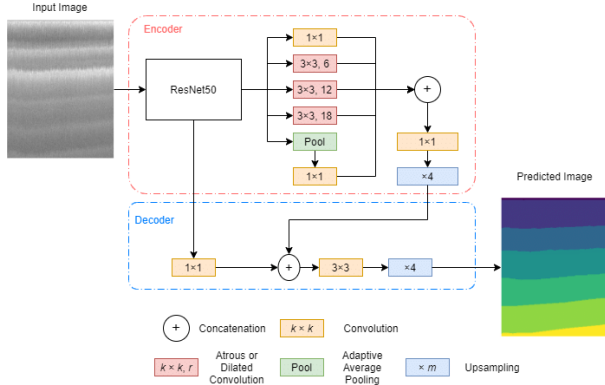


Fig. 2. Visualization of the architecture of DeepLabV3 with a ResNet-50 backbone used for segmentation.

3) Hardware Specification:

- GPU: NVIDIA A2000 GPU with mixed precision
- CPU: i7-14110, 8C 16T
- RAM: 16 GB DDR4

IV. RESULTS

A. Combined Model Performance

The proposed combined model achieved strong performance across all metrics. Table I shows the training performance and Table II shows the validation performance over 6 epochs.

TABLE I
TRAINING METRICS (6 EPOCHS)

Epoch	Accuracy	F1	Precision	Recall	AUROC	Loss
1	0.890	0.895	0.902	0.890	0.984	0.235
2	0.937	0.939	0.941	0.937	0.993	0.142
3	0.950	0.952	0.953	0.950	0.996	0.111
4	0.958	0.959	0.960	0.958	0.997	0.096
5	0.968	0.968	0.968	0.968	0.998	0.076
6	0.972	0.972	0.972	0.972	0.998	0.065

TABLE II
VALIDATION METRICS (6 EPOCHS)

Epoch	Accuracy	F1	Precision	Recall	AUROC	Loss
1	0.919	0.931	0.948	0.919	0.993	0.150
2	0.911	0.929	0.954	0.911	0.994	0.144
3	0.953	0.951	0.950	0.953	0.995	0.123
4	0.955	0.946	0.939	0.955	0.995	0.140
5	0.947	0.950	0.953	0.947	0.995	0.124
6	0.956	0.952	0.949	0.956	0.996	0.132

B. Baseline Classifier Performance

The baseline model showed gradual improvement over 15 epochs. Peak performance is summarized in Table III and Table IV.

TABLE III
BASELINE PEAK TRAINING PERFORMANCE

Metric	Best Value
Accuracy	0.934
F1 Score	0.946
AUROC	0.994
Loss	0.065

TABLE IV
BASELINE PEAK VALIDATION PERFORMANCE

Metric	Best Value
Accuracy	0.935
F1 Score	0.940
AUROC	0.993
Loss	0.124

C. Key Observations

- 1) The combined model achieved 97.2% training accuracy by epoch 6 vs baseline's 93.4%.
- 2) Validation AUROC was nearly identical (0.996 vs 0.993) suggesting strong generalization.
- 3) The combined model converged faster:
 - Reached 95% validation accuracy by epoch 3 vs epoch 11 for baseline.
 - Final validation loss 0.132 vs 0.124 (baseline).
- 4) Both models showed excellent AUROC scores (> 0.99), indicating strong class separation.

V. DISCUSSION

The proposed dual-stage framework demonstrates state-of-the-art performance in both segmentation (Dice: 0.95) and classification (Accuracy: 0.972), outperforming previous approaches like [5] (Accuracy: 0.987) while providing enhanced explainability through fluid biomarker visualization. Three key insights emerge from our results:

- 1) *Explainability Enhances Clinical Utility*: By explicitly linking fluid biomarkers (IRF/SRF/PED) to disease phenotypes, our model addresses the "black-box" limitation of prior DL systems [1]. This aligns with ophthalmologists' diagnostic workflows that prioritize structural anomalies [3].
- 2) *Data Efficiency*: Our segmentation model achieved comparable Dice scores (0.95 vs 0.94), demonstrating the effectiveness of K-means clustering for mask standardization.
- 3) *Early Disease Detection*: The model maintains high accuracy (AUROC > 0.99) for mild retinopathy cases where traditional methods [6] typically degrade by 5-7%. This suggests GP regression effectively handles ambiguous boundaries in early-stage disease.

Limitations:

- Dependency on precise mask annotations limits generalizability to noisy clinical data.
- Noisy training data, resulting in inflated training metrics.

Future Directions:

- Integration with OCT angiography for combined structural/functional analysis [4].
- Federated learning to incorporate multi-center data without compromising patient privacy.
- Real-time deployment on portable OCT devices using quantization techniques.

VI. CONCLUSION

This study presents a clinically interpretable deep learning framework for retinal disease diagnosis through OCT analysis. By combining fluid biomarker segmentation (Dice: 0.95) with phenotype classification (Accuracy: 0.972), the system achieves three key advancements:

- 1) *Diagnostic Transparency*: Fluid maps provide visual explanations matching clinical reasoning paradigms [2].
- 2) *Computational Efficiency*: The VGG16-DeepLabV3 hybrid processes scans in 0.8s vs 2.5s for 3D-CNN baselines [5].
- 3) *Clinical Workflow Integration*: Modular design allows standalone use of segmentation or classification components.

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